ACTIVITY 1

Find out the symbols of the elements with the help of your teacher.

Elements	Symbol
Gold	
Silver	
Copper	
Iron	
Nitrogen	
Oxygen	
Aluminium	
Calcium	
Phosphorus	
Magnesium	
Potassium	
Sodium	

- Chemical symbols usually consist of one or two letters.
- The symbols of most elements correspond to the first letter (which is capitalized) of their English name. For example, the symbol for oxygen is O and that for hydrogen is H. You will study about symbols in details in standard 8.

3.3.3 Elements in human Body

Nearly 99% of the mass of our human body consists of just six chemical elements namely, oxygen, carbon, hydrogen, nitrogen, calcium, and phosphorus. Another five elements make up most of the least percentage. They are potassium, sulphur, sodium, chlorine, and magnesium.

3.3.4 Elements in air

Air is a mixture of gases. The molecules of two different elements, nitrogen and oxygen, make up about 99% of the air. The rest includes small amounts of argon and carbon dioxide. Other gases such as neon, helium, and methane are present in trace amounts. Oxygen is the life-giving element in the air.

3.4 Compounds

A compound is a pure substance that is formed when the atoms of two or more elements combine chemically in definite proportions.

Compounds exhibit properties that are entirely different from the properties of their constituent elements. For example, the atoms of the elements hydrogen and oxygen combine chemically in a fixed ratio to form the compound water. However, water does not have the same properties of hydrogen and oxygen. For example, at room temperature water exists as liquid while hydrogen and oxygen exist as gases. Also, oxygen supports fire whereas water is used as a fire extinguisher.

Similarly, common salt (Sodium chloride) is a compound made up of elements sodium and chlorine. It is used in our food, whereas sodium and chlorine are poison, and both are unsafe for consumption.



Sodium is a highly reactive solid at room temperature . It burns vigorously when in contact with water

+



Chlorine is yellowish green poisonous gas at room temperature

→



Sodium Chloride (Used for cooking)

VII Std Science Term-1 EM Unit 3.indd 33 24-03-2020 14:18:52